

Ali Aqdas

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RESEARCH INTERESTS

Systems Security, Microarchitectural Security, Disaggregated Memory (CXL), Hardware-Software Co-design, and RISC-V Architecture.

TECHNICAL EXPERTISE

Languages	C, C++, Python, SystemVerilog, x86-64 / ARM64 / RISC-V Assembly, MATLAB
Systems	Cache Coherence (MESI/MESIF), CXL 2.0/3.0, TEEs, RDMA (CoyoteOS), Linux Kernel
Security	Side-Channel Attacks, Transient Execution, Ghidra, ROPgadget, pwntools
Hardware/RTL	FPGA (Xilinx/Intel), ASIC Flow (TSMC 22/65nm, ASAP7), Genus, Innovus, Verilator

EDUCATION

Purdue University	Aug 2024 – Present
Ph.D. in Computer Science	West Lafayette, IN
Advisor: Dr. Kazem Taram	
Research: Microarchitectural Security in Compute Express Link (CXL) Systems	
National University of Sciences and Technology (NUST)	Sept 2019 – Jun 2023
Bachelor of Electrical Engineering	Islamabad, Pakistan
CGPA: 3.93/4.00	
Thesis: SoC Implementation of RISC-V Vector Processor (RVV 1.0)	

PUBLICATIONS

Ali Aqdas, Zheng Zhang, Yue Pan, Zixuan Wang, Tajana Rosing, Jishen Zhao, Kazem Taram.
"Compute eXpress Leak: New Cache Attacks Using CXL Devices."
Under review at IEEE/ACM MICRO 2026.

RESEARCH EXPERIENCE

SecArch Lab, Purdue University	Mar 2025 – Present
Graduate Research Assistant	West Lafayette, IN
- Discovered novel protocol-level vulnerabilities in CXL cache coherence, exposing the first cross-host and device-initiated side-channel attacks. - Designed and implemented PUSH+PUSH and SnoopDog attacks, leveraging CXL-native coherence messages to leak data across host boundaries. - Developed a full-stack hardware/software platform using Linux drivers and FPGA-based CXL RTL to demonstrate covert channels at ~17 kbps with < 9% error. - Engineered a constant-time protocol mitigation for the CXL interface, maintaining high security with negligible (< 3.13%) performance overhead on physical hardware.	
Next-G Architectures Lab, Purdue University	Aug 2024 – Mar 2025
Graduate Research Assistant	West Lafayette, IN
Architected a high-throughput RDMA-based transport simulation by integrating CoyoteOS with SDNet for disaggregated accelerator clusters.	

- Optimized **hardware-accelerated transport layers** to reduce protocol overhead for unreliable communication in distributed ML training workloads.
- Conducted hardware/software co-design evaluations to identify and mitigate memory subsystem bottlenecks in large-scale distributed architectures.

SoC Design Lab, NUST

Research Assistant

Jun 2022 – Jul 2024

Islamabad, Pakistan

- Led the **RTL design and integration** of a RISC-V Vector Processor (RVV 1.0) with a custom scalar core.
- Executed the complete **ASIC synthesis and floorplanning flow** (TSMC 65nm/22nm, ASAP7) using Cadence Genus and Innovus.
- Analyzed **PPA tradeoffs** (Power, Area, and Timing) and optimized microarchitectural components for high-bandwidth vector operations.
- Developed vectorized kernels using intrinsic instructions for performance benchmarking and architectural verification.

TEACHING EXPERIENCE

Purdue University

Graduate Teaching Assistant

Aug 2024 – Present

West Lafayette, IN

- **CS-250: Computer Architecture** (Fall 2024, Spring 2025, Fall 2025).
- Responsible for leading lab sessions, developing course materials, and assisting students with low-level systems programming and architectural concepts.

NUST Chip Design Center

Instructor

May 2024 – July 2024

Islamabad, Pakistan

- Designed and conducted hands-on technical labs for training industry recruits in ASIC design flows.
- Collaborated with faculty to develop advanced training modules for RISC-V microarchitecture.

SERVICE & HONORS

Artifact Evaluator, ISCA 2025–2026: Evaluated reproducibility and technical rigor of research artifacts for the International Symposium on Computer Architecture.

NUST Scholarship Award: Received for top academic performance across all four years of undergraduate study (2019–2023).

Academic Scholarship: Awarded for excellence in Higher Secondary and Secondary Education (2015–2019).

SELECTED PROJECTS

XOR-Memory Generator: Developed a software tool in C++ to generate parameterizable Verilog files for multi-ported memory systems on FPGAs, enabling high-bandwidth parallel access.

TinyML Fall Detection System: Engineered and deployed an optimized CNN using TensorFlow Lite Micro on ARM Cortex-M microcontrollers for real-time edge inference.

Bresenham Circle Algorithm on FPGA: Implemented circle-drawing logic in Verilog with VGA output on a DE1-SoC platform.

Angle of Arrival Estimation: Simulated the MUSIC algorithm in MATLAB and implemented it on FPGA for real-time signal processing in the 4–6 GHz range.